

Sadık Bera Yüksel

yuksel.sa@northeastern.edu | 857-991-4390 | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#) | Boston, MA

EDUCATION

Northeastern University, Boston, MA

PhD in Computer Engineering | GPA:4.0/4.0

Expected May 2028

Relevant Coursework: Robotic Sensing and Navigation, Reinforcement Learning, Mobile Robotics, Advances in Deep Learning, Verifiable Machine Learning, Linear System Analysis, Formal Methods for Dynamical Systems

Bilkent University, Ankara, Turkey

Bachelor of Science, Electrical and Electronics Engineering

June 2023

Relevant Coursework: Introduction to Machine Learning, Microprocessors, Natural Language Processing

Korea University, (Seoul, South Korea)

Study Abroad Program

February - June 2022

Relevant Coursework: Statistical Modeling for Deep Learning, Computer System Design, Artificial Intelligence

Honors: Global KU Scholarship

EXPERIENCE

Northeastern University, Boston, MA

Graduate Research Assistant (Advisor: Prof. Derya Aksaray)

September 2023 – Present

- Conducting research on temporal-logic-constrained reinforcement learning that incorporates STL specifications as hard constraints to enhance the safety and real-world applicability of reinforcement learning.
- Developed a temporal-logic-guided motion planning framework with contingency constraints to ensure robustness of autonomous systems operating under uncertainty.
- Deployed the planning algorithm on a real drone in a GPS-denied environment, integrating perception, control, and runtime monitoring for autonomous task execution.

ASELSAN, Ankara, Turkey

Transmitter and Receiver Design Intern

July - August 2022

- Designed a diplexer consisting of two RF bandpass filters with distinct operation bands to improve signal routing efficiency using Keysight Genesys software.
- Created PCB layouts for RF filters and manufactured the designed diplexer with lumped elements.
- Analyzed the manufactured diplexer's S-parameters with a vector network analyzer to verify compliance with requirements.

HAVELSAN, Ankara, Turkey

Robotics and Autonomous Systems Intern

June - August 2021

- Implemented autonomous navigation for a differential drive robot in a simulation environment using ROS and Python.
- Developed a face recognition application on Qualcomm Robotics RB5 using OpenCV and optimized the pipeline for real-time inference on embedded hardware.

Bilkent University, Ankara, Turkey

Teaching Assistant

September 2019 - January 2020

- Assisted students in the CS115 - Introduction to Programming in Python course during lab hours, providing targeted educational support that improved student understanding and engagement in programming concepts.

PUBLICATIONS

- X. Lin, **S.B. Yüksel**, Y. Yazıcıoğlu, and D. Aksaray, "Probabilistic Satisfaction of Temporal Logic Constraints in Reinforcement Learning via Adaptive Policy-Switching", *Proc. of the Learning for Dynamics & Control (L4DC)*, 2025.
- **S.B. Yüksel**, A. Taheri, Y. Yazıcıoğlu, and D. Aksaray, "Robust Temporal Logic Planning Under Contingency Constraints", *International Conference on Automation Science and Engineering (CASE)*, 2025.
- **S.B. Yüksel**, A.T. Buyukkocak, and D. Aksaray, "Shielded Reinforcement Learning Under Dynamic Temporal Logic Constraints" (Under review), 2025.

SKILLS

- **Programming Languages:** Python, MATLAB, Assembly
- **Machine Learning & AI:** PyTorch, JAX, TensorFlow, NumPy, Pandas, Scikit-learn, OpenCV
- **Robotics & Control:** ROS, SLAM, motion planning, reinforcement learning, control systems
- **Tools and Software:** Git, Docker, Linux, LTspice, Vivado, Diptrace, Proteus 8

ACADEMIC PROJECTS

Sketch-to-Image Translation for Human Identification, Northeastern University Spring 2025

- Implemented a sketch-to-image GAN in PyTorch to reconstruct realistic faces from sketch inputs for identification tasks.

Visual SLAM Using Real Car Data, Northeastern University Fall 2023

- Collected real car IMU and depth camera data and implemented a visual SLAM algorithm in ROS to generate accurate real-world maps.

Indoors Drone Localization Using UWB Sensors, Bilkent University Spring 2023

- Developed an indoor drone localization system in Python that used Ultra-Wideband (UWB) distance data with trilateration and filtering to determine real-time position estimates.
- Deployed the algorithm on the drone's onboard computer and created a real-time monitoring interface for ground-based visualization.